

An Inexpensive Gouy Balance for Magnetic Susceptibility Determination

In many a school and college double pan equal arm balances are becoming obsolete and are being rapidly replaced by automatic and semiautomatic single pan balances and digital balances. Instead of discarding a double pan balance, in conjunction with a permanent magnet, can be converted into a Gouy balance. The student is introduced to the principles and operation of a double pan balance¹ and magnetic properties of materials² simultaneously through a single experiment.

A wooden desk (107 × 66 × 76 cm) is converted into the magnet housing by covering the three sides with 3-mm thick plywood. The fourth side is provided with a plywood door that can be opened or closed at will by sliding it on guide slots. A 6-mm diam hole is drilled on the desk top to introduce the suspension system into the magnetic field. The magnet is an already available Master Magnet No. 59 × 64 supplied by Cambosco Scientific Company, Brighton Station, Mass. The magnet has pole faces of ~30 mm and a pole gap of about the same dimensions. The magnet has a flux density of 3000 Gauss. A magnet of similar flux density and variable pole gap is obtainable from Edmund Scientific Company (Cat. No. 70,810) for under \$60. The magnet is mounted on the foot rest of the desk with a 102-mm door hinge so that the position of the magnet can be varied to facilitate weighing of the sample in and out of field. Only minor alterations are made to the chainomatic balance. A 0.5-mm diam hole is drilled at the center of the left pan. The pan support for the left pan is disconnected and a 6-mm diam hole is drilled into the base of the balance to insert the suspension system carrying the sample tube. The balance is mounted on the desk top such that the center of the left pan from

which the sample tube is suspended is directly above the center of the magnetic field. Thus, the sample tube when suspended will be located exactly between the poles of the magnet. The sample tube consists of an 80-mm long and 6-mm o.d. Pyrex tube the bottom 20 mm of which are sealed with a low melting alloy. The top of the tube is flared in such a manner that a harness can be attached. The tube is connected to one end of a Nylon cord with a silver hook to facilitate easy removal of the tube for sample loading or cleaning. The other end of the cord is passed through the holes in the desk top, balance base, and pan, and is fastened with a knot.

Students are directed to weigh the samples by the rest point method, determine the magnetic susceptibility, and calculate the number of unpaired electrons in the molecule. Magnetic susceptibilities of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, $\text{Ni}(\text{NO}_3)_2$, chrome alum, $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$, and $\text{MnSO}_4 \cdot \text{H}_2\text{O}$ measured with this instrument were, respectively, 2.01, 3.13, 3.84, 4.98, and 5.92 BM and compare well with the expected values for these compounds.³

We have found that this experiment is effective in elucidating the principles of operation of a double pan analytical balance and introducing the student to the magnetic properties of materials. With the availability of a desk and an analytical balance the cost of construction of a Gouy balance is well under \$100. and can be fabricated in a relatively short time.

¹Evans, L., J. CHEM. EDUC., **32**, 419 (1955).

²Barrow, G. M., "Physical Chemistry," 2nd Ed., McGraw-Hill Book Co., New York, 1966, p. 438.

³Figgis, B. N., and Lewis, J., "Modern Coordination Chemistry," (Editors: Lewis, J., and Wilkins, R. J.), Interscience, New York, 1960, p. 406.